

The Theory That Men Evolved to Hunt and Women Evolved to Gather Is Wrong

The influential idea that in the past men were hunters and women were not isn't supported by the available evidence

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Credit: Samantha Mash

Even if you're not an anthropologist, you've probably encountered one of this field's most influential notions, known as Man the Hunter. The theory proposes that hunting was a major driver of human evolution and that men carried this activity out to the exclusion of women. It holds that human ancestors had a division of labor, rooted in biological differences between males and females, in which males evolved to hunt and provide, and females tended to children and domestic duties. It assumes that males are physically superior to females and that pregnancy and child-rearing reduce or eliminate a female's ability to hunt.

Man the Hunter has dominated the study of human evolution for nearly half a century and pervaded popular culture. It is represented in museum dioramas and textbook figures, Saturday morning cartoons and feature films. The thing is, it's wrong.

Mounting evidence from exercise science indicates that women are physiologically better suited than men to endurance efforts such as running marathons. This advantage bears on questions about hunting because a prominent hypothesis contends that early humans are thought to have pursued prey on foot over long distances until the animals were exhausted. Furthermore, the fossil and archaeological records, as well as ethnographic studies of modern-day hunter-gatherers, indicate that women have a long history of hunting game. We still have much to learn about female athletic performance and the lives of prehistoric women. Nevertheless, the data we do have signal that it is time to bury Man the Hunter for good.

The theory rose to prominence in 1968, when anthropologists Richard B. Lee and Irvén DeVore published *Man the Hunter*, an edited collection of scholarly papers presented at a 1966 symposium on contemporary hunter-gatherer societies. The volume drew on ethnographic, archaeological and paleoanthropological evidence to argue that hunting is what drove human evolution and resulted in our suite of unique features. “Man's life as a hunter supplied all the other ingredients for achieving civilization: the genetic variability, the inventiveness, the systems of vocal communication, the coordination of social life,” anthropologist William S. Laughlin writes in chapter 33 of the book. Because men were supposedly the ones hunting, proponents of the Man the Hunter theory assumed evolution was acting primarily on men, and women were merely passive beneficiaries of both the meat supply and evolutionary progress.

But *Man the Hunter's* contributors often ignored evidence, sometimes in their own data, that countered their suppositions. For example, Hitoshi Watanabe focused on ethnographic data about the Ainu, an Indigenous population in northern Japan and its surrounding areas. Although Watanabe documented Ainu women hunting, often with the aid of dogs, he dismissed this finding in his interpretations and placed the focus squarely on men as the primary meat winners. He was superimposing the idea of male superiority through hunting onto the Ainu and into the past.

This fixation on male superiority was a sign of the times not just in academia but in society at large. In 1967, the year between the *Man the Hunter* conference and the publication of the edited volume, 20-year-old Kathrine Switzer entered the Boston Marathon under the name “K. V. Switzer,” which obscured her gender. There were no

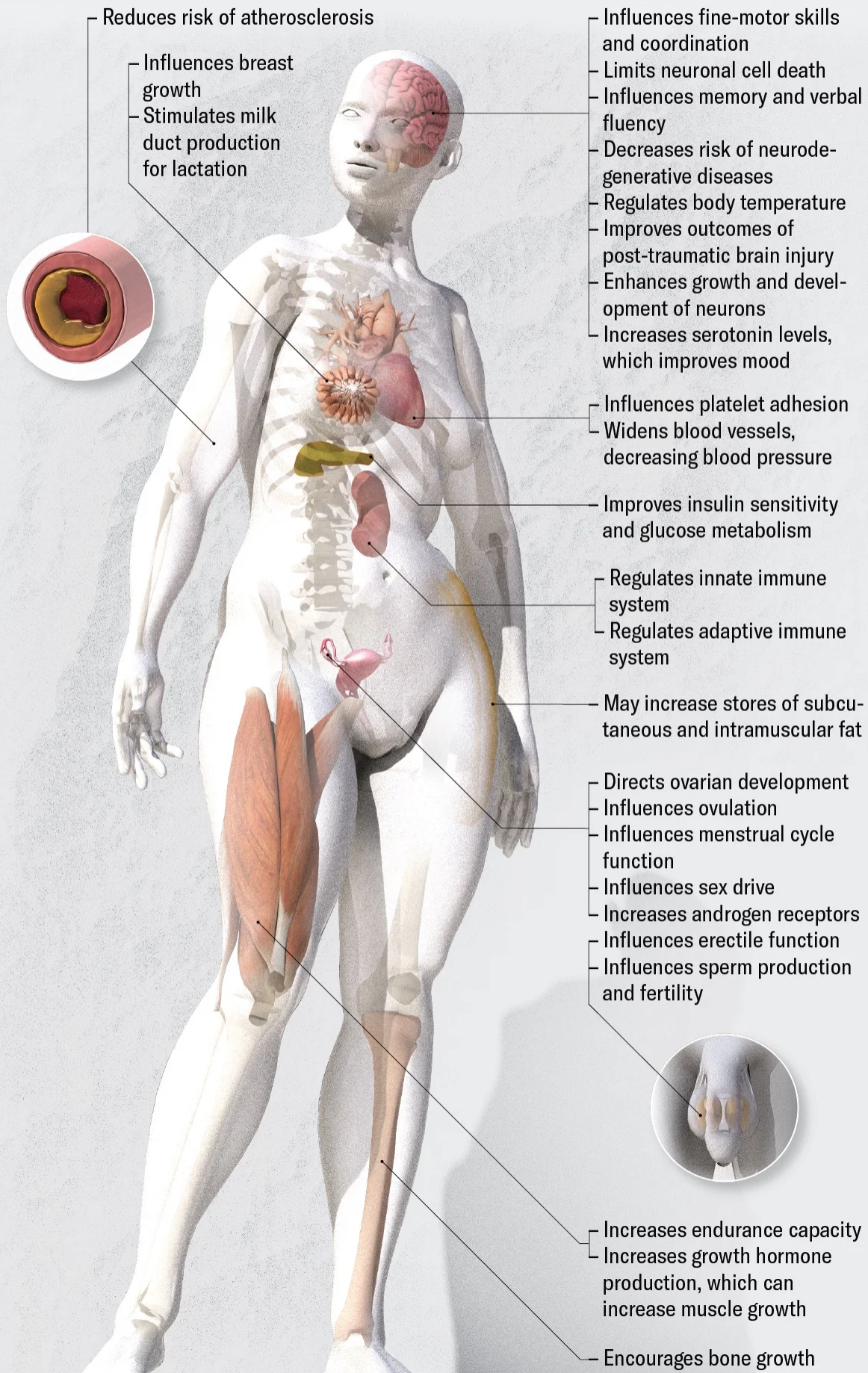
official rules against women entering the race; it just was not done. When officials discovered that Switzer was a woman, race manager Jock Semple attempted to push her physically off the course.

At that time, the conventional wisdom was that women were incapable of completing such a physically demanding task and that attempting to do so could harm their precious reproductive capacities. Scholars following Man the Hunter dogma relied on this belief in women's limited physical capacities and the assumed burden of pregnancy and lactation to argue that only men hunted. Women had children to rear instead.

Today these biased assumptions persist in both the scientific literature and the public consciousness. Granted, women have recently been shown hunting in movies such as *Prey*, the most recent installment of the popular Predator franchise, and on cable programs such as *Naked and Afraid* and *Women Who Hunt*. But social media trolls have viciously critiqued and labeled these depictions as part of a politically correct feminist agenda. They insist the creators of such works are trying to rewrite gender roles and evolutionary history in an attempt to co-opt “traditionally masculine” social spheres. Bystanders might be left wondering whether portrayals of women hunters are trying to make the past more inclusive than it really was—or whether Man the Hunter–style assumptions about the past are attempts to project sexism backward in time. Our recent surveys of the physiological and archaeological evidence for hunting capability and sexual division of labor in human evolution answer this question.

Powers of Estrogen

The hormone testosterone gets all the attention in the fitness world. But the hormone estrogen, which females typically produce more of than males, actually plays a critical role in athletic performance. In addition to regulating the reproductive systems of both males and females, estrogen exerts effects throughout the body.





Credit: Violet Isabelle Frances for Bryan Christie Design

Before getting into the evidence, we need to first talk about sex and gender. “Sex” typically refers to biological sex, which can be defined by myriad characteristics such as chromosomes, hormone levels, gonads, external genitalia and secondary sex characteristics. The terms “female” and “male” are often used in relation to biological sex. “Gender” refers to how an individual identifies—woman, man, nonbinary, and so forth. Much of the scientific literature confuses and conflates female/male and woman/man terminology without providing definitions to clarify what it is referring to and why those terms were chosen. For the purpose of describing anatomical and physiological evidence, most of the literature uses “female” and “male,” so we use those words here when discussing the results of such studies. For ethnographic and archaeological evidence, we are attempting to reconstruct social roles, for which the terms “woman” and “man” are usually used. Unfortunately, both these word sets assume a binary, which does not exist biologically, psychologically or socially. Sex and gender both exist as a spectrum, but when citing the work of others, it is difficult to add that nuance.

It also bears mentioning that much of the research into exercise physiology, paleoanthropology, archaeology and ethnography has historically been conducted by men and focused on males. For example, Ella Smith of the Australian Catholic University and her colleagues found that in studies of nutrition and supplements, only 23 percent of participants were female. In studies focusing on athletic performance, Emma Cowley of the University of North Carolina at Chapel Hill and her colleagues found, only 3 percent of publications had female-only participants; 63 percent of publications looked exclusively at males. This massive disparity means we still know very little about female athletic performance, training and nutrition, leaving athletic trainers and coaches to mostly treat females as small males. It also means that much of the work we have to rely on to make our physiological arguments about female hunters in prehistory is based on research with small human sample sizes or rodent studies. We hope this state of affairs will inspire the next generation of scientists to ensure that females are represented in such studies. But even with the limited data available to us, we can show that Man the Hunter is a flawed theory and make the case that females in early human communities hunted, too.

From a biological standpoint, there are undeniable differences between females and males. When we discuss these differences, we are typically referring to means,

averages of one group compared with another. Means obscure the vast range of variation in humans. For instance, although males tend to be larger and to have bigger hearts and lungs and more muscle mass, there are plenty of females who fall within the typical male range; the inverse is also true.

Overall, females are metabolically better suited for endurance activities, whereas males excel at short, powerful burst-type activities. You can think of it as marathoners (females) versus powerlifters (males). Much of this difference seems to be driven by the powers of the hormone estrogen.

Female vs. Male Athletic Advantages

Females and males differ biologically in ways that translate to different athletic advantages. Females are better able to use fat for sustained energy and keep their muscles in better condition during exercise, for instance—traits that give them an advantage in endurance activities. In light of this physiological evidence, along with archaeological evidence, it stands to reason that females in early human communities hunted, just as females in later hunter-gatherer societies such as the Ainu of Japan have traditionally done.

THE SET OF FEMALE-ASSOCIATED FEATURES THAT CONFER AN ENDURANCE ACTIVITY ADVANTAGE

Better insulin sensitivity helps to prevent muscle breakdown during exercise by increasing fat burn

Wider pelvis may be more efficient for carrying hip-placed load

Greater fat stores aid endurance

Higher estrogen levels improve athletic performance by:

- Increasing fatty acid oxidation
- Decreasing glycogen utilization
- Increasing insulin sensitivity
- Sparing protein
- Attenuating heat-shock protein response
- Improving cellular membrane stabilization during stress
- Increasing the number of androgen receptors
- Increasing growth hormone, which can increase muscle mass
- Improving muscle recovery

Better psychological pacing
Greater fatigue resistance through central and peripheral neuromuscular mechanisms

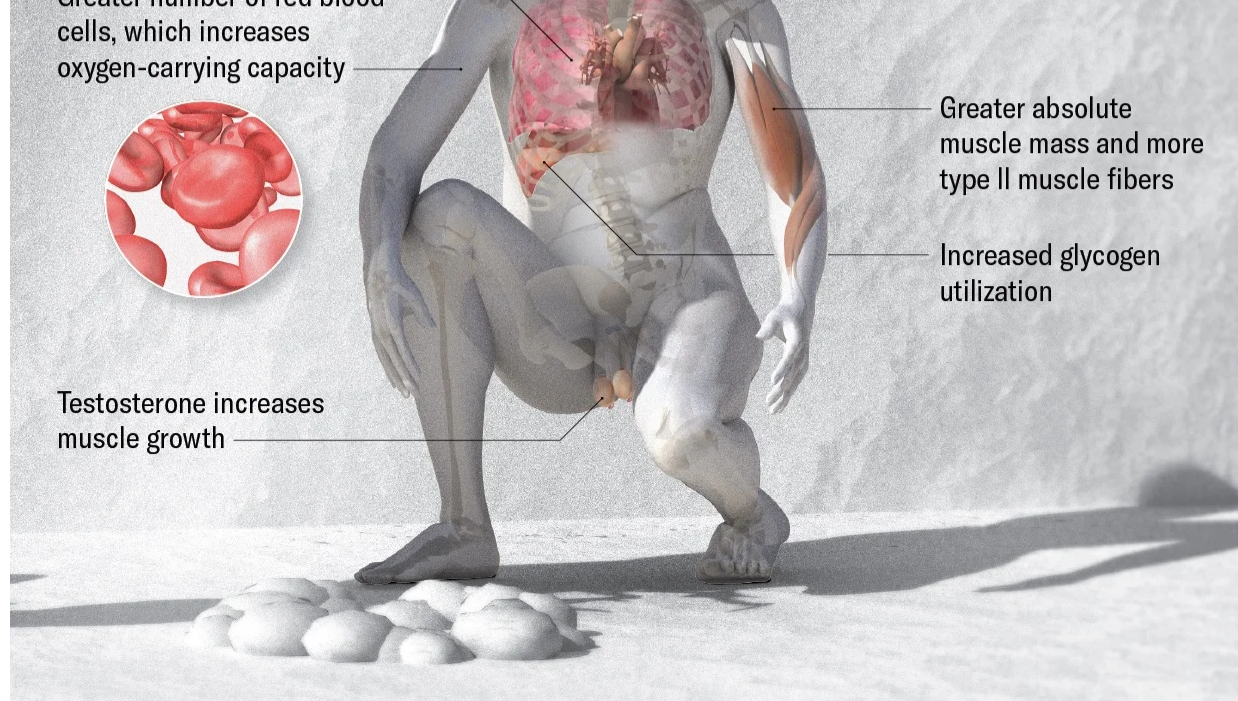
More type I muscle fibers increase endurance
Greater intramuscular fat stores increase energy availability
More effective stretch-shorten cycles make locomotion more efficient

THE SET OF MALE-ASSOCIATED FEATURES THAT CONFER A POWER ACTIVITY ADVANTAGE

Larger heart and lungs

Greater number of red blood





Credit: Violet Isabelle Frances for Bryan Christie Design

Given the fitness world's persistent touting of the hormone testosterone for athletic success, you'd be forgiven for not knowing that estrogen, which females typically produce more of than males, plays an incredibly important role in athletic performance. It makes sense from an evolutionary standpoint, however. The estrogen receptor—the protein that estrogen binds to in order to do its work—is deeply ancient. Joseph Thornton of the University of Chicago and his colleagues have estimated that it is around 1.2 billion to 600 million years old—roughly twice as old as the testosterone receptor. In addition to helping regulate the reproductive system, estrogen influences fine-motor control and memory, enhances the growth and development of neurons, and helps to prevent hardening of the arteries.

Important for the purposes of this discussion, estrogen also improves fat metabolism. During exercise, estrogen seems to encourage the body to use stored fat for energy before stored carbohydrates. Fat contains more calories per gram than carbohydrates do, so it burns more slowly, which can delay fatigue during endurance activity. Not only does estrogen encourage fat burning, but it also promotes greater fat storage within muscles—marbling if you will—which makes that fat's energy more readily available. Adiponectin, another hormone that is typically present in higher amounts in females than in males, further enhances fat metabolism while sparing carbohydrates for future use, and it protects muscle from breakdown. Anne Friedlander of Stanford University and her colleagues found that females use as much as 70 percent more fat for energy during exercise than males.

Correspondingly, the muscle fibers of females differ from those of males. Females have more type I, or “slow-twitch,” muscle fibers than males do. These fibers generate energy slowly by using fat. They are not all that powerful, but they take a long time to become fatigued. They are the endurance muscle fibers. Males, in contrast, typically have more type II (“fast-twitch”) fibers, which use carbohydrates to provide quick energy and a great deal of power but tire rapidly.

Females also tend to have a greater number of estrogen receptors on their skeletal muscles compared with males. This arrangement makes these muscles more sensitive to estrogen, including to its protective effect after physical activity. Estrogen's ability to increase fat metabolism and regulate the body's response to the hormone insulin can help prevent muscle breakdown during intense exercise. Furthermore, estrogen appears to have a stabilizing effect on cell membranes that might otherwise rupture from acute stress brought on by heat and exercise. Ruptured cells release enzymes called creatine kinases, which can damage tissues.

Studies of females and males during and after exercise bolster these claims. Linda Lamont of the University of Rhode Island and her colleagues, as well as Michael Riddell of York University in Canada and his colleagues, found that females experienced less muscle breakdown than males after the same bouts of exercise. Tellingly, in a separate study Mazen J. Hamadeh of York University and his colleagues found that males supplemented with estrogen suffered less muscle breakdown during cycling than those who didn't receive estrogen supplements. In a similar vein, research led by Ron Maughan of the University of St Andrews in Scotland found that females were able to perform significantly more weight-lifting repetitions than males at the same percentages of their maximal strength.

If females are better able to use fat for sustained energy and keep their muscles in better condition during exercise, then they should be able to run greater distances with less fatigue relative to males. In fact, an analysis of marathons carried out by Robert Deaner of Grand Valley State University demonstrated that females tend to slow down less as the race progresses compared with males.

If you follow long-distance races, you might be thinking, wait—males are outperforming females in endurance events! But this is only sometimes the case. Females are more regularly dominating ultraendurance events such as the more than 260-mile Montane Spine foot race through England and Scotland, the 21-mile swim across the English Channel and the 4,300-mile Trans Am cycling race across the U.S. Sometimes female athletes compete in these races while attending to the needs of their

children. In 2018 English runner Sophie Power ran the 105-mile Ultra-Trail du Mont-Blanc race in the Alps while still breastfeeding her three-month-old at rest stations.

The inequity between male and female athletes is a result not of inherent biological differences between the sexes but of biases in how they are treated in sports. As an example, some endurance-running events allow the use of professional runners called pacesetters to help competitors perform their best. Men are not permitted to act as pacesetters in many women's events because of the belief that they will make the women “artificially faster,” as though women were not actually doing the running themselves.

The modern physiological evidence, along with historical examples, exposes deep flaws in the idea that physical inferiority prevented females from partaking in hunting during our evolutionary past. The evidence from prehistory further undermines this notion.

Consider the skeletal remains of ancient people. Differences in body size between females and males of a species, a phenomenon called sexual size dimorphism, correlate with social structure. In species with pronounced size dimorphism, larger males compete with one another for access to females, and among the great apes larger males socially dominate females. Low sexual size dimorphism is characteristic of egalitarian and monogamous species. Modern humans have low sexual size dimorphism compared with the other great apes. The same goes for human ancestors spanning the past two million years, suggesting that the social structure of humans changed from that of our chimpanzeelike ancestors.



Anthropologists also look at damage on our ancestors' skeletons for clues to their behavior. Neandertals are the best-studied extinct members of the human family because we have a rich fossil record of their remains. Neandertal females and males do not differ in their trauma patterns, nor do they exhibit sex differences in pathology from repetitive actions. Their skeletons show the same patterns of wear and tear. This finding suggests that they were doing the same things, from ambush-hunting large game animals to processing hides for leather. Yes, Neandertal women were spearing woolly rhinoceroses, and Neandertal men were making clothing.

Males living in the Upper Paleolithic—the cultural period between roughly 45,000 and 10,000 years ago, when early modern humans entered Europe—do show higher rates of a set of injuries to the right elbow region known as thrower's elbow, which could mean they were more likely than females to throw spears. But it does not mean women were not hunting, because this period is also when people invented the bow and arrow, hunting nets and fishing hooks. These more sophisticated tools enabled humans to catch a wider variety of animals; they were also easier on hunters' bodies. Women may have favored hunting tactics that took advantage of these new technologies.

What is more, females and males were buried in the same way in the Upper Paleolithic. Their bodies were interred with the same kinds of artifacts, or grave goods, suggesting that the groups they lived in did not have social hierarchies based on sex.

Ancient DNA provides additional clues about social structure and potential gender roles in ancestral human communities. Patterns of variation in the Y chromosome, which is paternally inherited, and in mitochondrial DNA, which is maternally inherited, can reveal differences in how males and females dispersed after reaching maturity. Thanks to analyses of DNA extracted from fossils, we now know of three Neandertal groups that engaged in patrilocality—wherein males were more likely to stay in the group they were born into and females moved to other groups—although we do not know how widespread this practice was.

Patrilocality is believed to have been an attempt to avoid incest by trading potential mates with other groups. Nevertheless, many Neandertals show both genetic and anatomical evidence of repeated inbreeding in their ancestry. They lived in small, nomadic groups with low population densities and endured frequent local extinctions, which produced much lower levels of genetic diversity than we see in living humans. This is probably why we don't see any evidence in their skeletons of sex-based

differences in behavior. For those practicing a foraging subsistence strategy in small family groups, flexibility and adaptability are much more important than rigid roles, gendered or otherwise. Individuals get injured or die, and the availability of animal and plant foods changes with the seasons. All group members need to be able to step into any role depending on the situation, whether that role is hunter or breeding partner.

Observations of recent and contemporary foraging societies provide direct evidence of women participating in hunting. The most cited examples come from the Agta people of the Philippines. Agta women hunt while menstruating, pregnant and breastfeeding, and they have the same hunting success as Agta men.

They are hardly alone. A recent study of ethnographic data spanning the past 100 years—much of which was ignored by *Man the Hunter* contributors—found that women from a wide range of cultures hunt animals for food. Abigail Anderson and Cara Wall-Scheffler of Seattle Pacific University and their colleagues report that 79 percent of the 63 foraging societies with clear descriptions of their hunting strategies feature women hunters. The women participate in hunting regardless of their childbearing status. These findings directly challenge the Man the Hunter assumption that women's bodies and childcare responsibilities limit their efforts to gathering foods that cannot run away.

So much about female exercise physiology and the lives of prehistoric women remains to be discovered. But the idea that in the past men were hunters and women were not is absolutely unsupported by the limited evidence we have. Female physiology is optimized for exactly the kinds of endurance activities involved in procuring game animals for food. And ancient women and men appear to have engaged in the same foraging activities rather than upholding a sex-based division of labor. It was the arrival some 10,000 years ago of agriculture, with its intensive investment in land, population growth and resultant clumped resources, that led to rigid gendered roles and economic inequality.

Now when you think of “cave people,” we hope, you will imagine a mixed-sex group of hunters encircling an errant reindeer or knapping stone tools together rather than a heavy-browed man with a club over one shoulder and a trailing bride. Hunting may have been remade as a masculine activity in recent times, but for most of human history, it belonged to everyone.

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